

Isoptopes and Minerals



Prof. A.K. Chaubey
akchaubey@iitism.ac.in

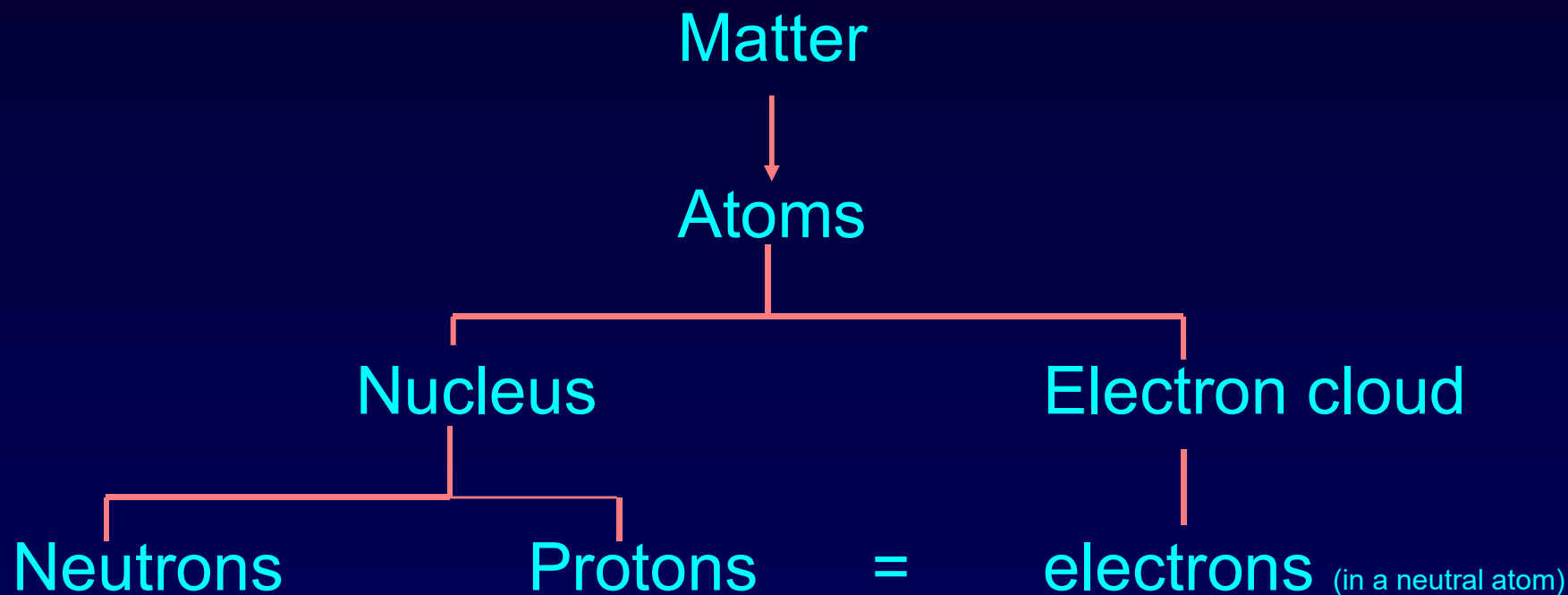
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Isoptopes

ATOM

- Atoms are the basic units of matter.
- The term "atom" comes from the Greek word for indivisible, because it was once thought that atoms were the smallest particle in the universe and could not be divided.
- We now know that atoms are made up of three particles known as subatomic particles: protons, neutrons and electrons - which are composed of even smaller particles, such as quarks.
- Atoms were created after the Big Bang ~14 billion years ago. As the hot, dense new universe cooled, conditions became suitable for quarks and electrons to form. Quarks came together to form protons and neutrons, and these particles combined into nuclei.
- An atom is composed of two entities: a central nucleus, containing most of the mass, and an array of orbiting electrons.

Atomic Structure



- The number of protons in an atom equals its atomic number
- Every atom has a unique atomic number
- Since electrons are easily gained or lost they determine the overall charge of an atom.

The nucleus was discovered in 1911 by Ernest Rutherford, a physicist from New Zealand. In 1920, Rutherford proposed the name proton for the positively charged particles of the atom. He also theorized that there was a neutral particle within the nucleus, which James Chadwick, a British physicist and student of Rutherford, was able to confirm in 1932.

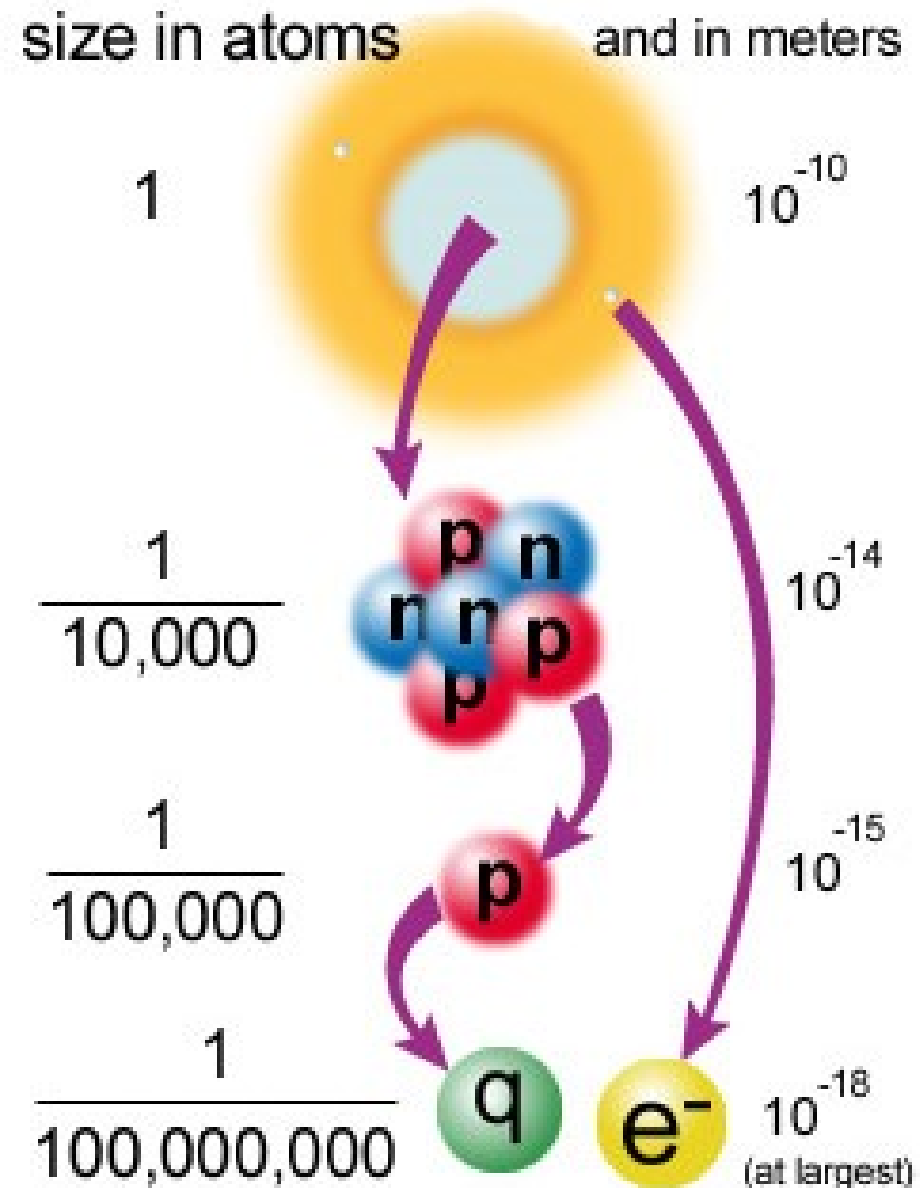
Atomic Structure

Particle	Symbol	Location	Charge	Mass
Proton	p^+	nucleus	1+	1 amu
Neutron	n^0	nucleus	0 No charge	1 amu
Electron	e^-	Electron cloud	1-	1/1840 amu

- Let's just consider the scale of the atom
- In fact the tiny electrons and quarks have no observed structure and are for all intents and purposes fundamental.

Charge and mass of the particles within atoms

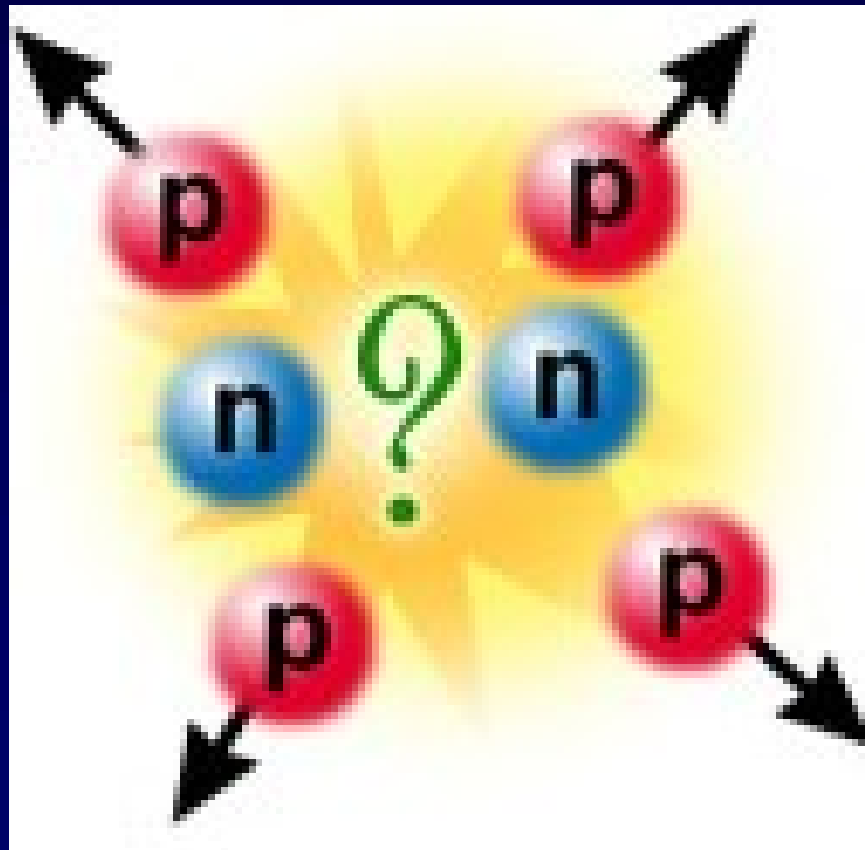
Elementary Particle	Charge	Mass
Proton	+1	1
Neutron	0	1
Electron	-1	~0



$$1\text{AMU} = 1.67 \times 10^{-27} \text{ kg}$$

The Problem of the Nucleus

- Why doesn't the nucleus - full of positive protons that repel one another and neutral neutrons - blow itself apart?



The Four Fundamental Forces

- The four fundamental forces: gravity, weak nuclear, electromagnetism, and strong nuclear which control the interactions in the universe
- All but gravity explained by the Standard Model of particle physics

Gravity: 1
Weak Nuclear Force : 10^{33}
Electromagnetic: 10^{36}
Strong Nuclear Force: 10^{38}
(Holds atomic nuclei together)

Gravity: infinite
Weak Nuclear Force : 10^{-18} m
Electromagnetic: infinite
Strong Nuclear Force: 10^{-15} m
(Holds atomic nuclei together)

Isotopes

- The term isotope is named from the Greek roots isos ("equal") and topos ("place"), meaning "the same place".
- Thus, the meaning behind the name is that different isotopes of a single element occupy the same position in the periodic table.
- A family of people often consists of related but not identical individuals. Elements have families as well, known as isotopes. Isotopes are members of a family of an element that all have the same number of protons but different numbers of neutrons.
- All isotopes of a given element have almost the same chemical properties, but they have different atomic masses and physical properties.
- The number of protons within the atom's nucleus is called atomic number and is equal to the number of electrons in the neutral (non-ionized) atom.

Isotopes

- Atoms with the same number of protons but different numbers of neutrons.
- An elements mass number is the number of protons plus the number of neutrons.
- To determine the number of neutrons subtract the atomic number from the mass number.
- $\text{Mass Number} - \text{atomic Number} = \text{Number of neutrons}$

Chemical elements

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra																
			La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
			Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Where do all the elements come from?

Were they formed at the same time?

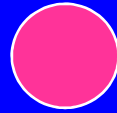
ATOMS

PROTON



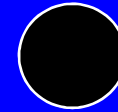
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NEUTRON



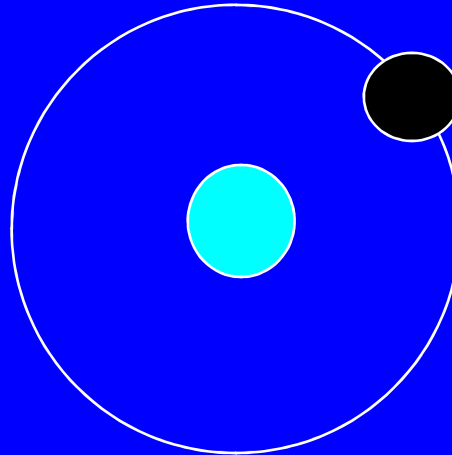
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ELECTRON

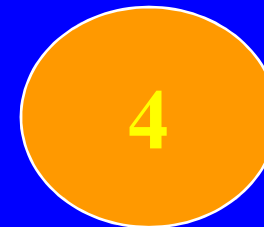
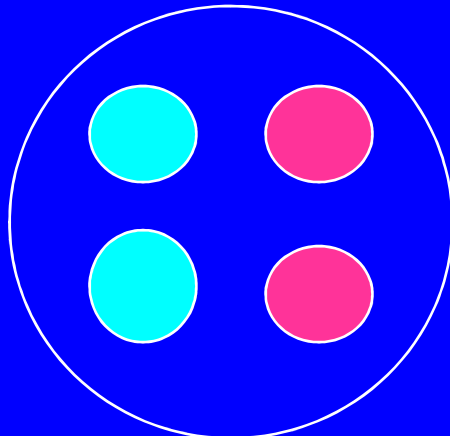


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HYDROGEN ATOM (H)

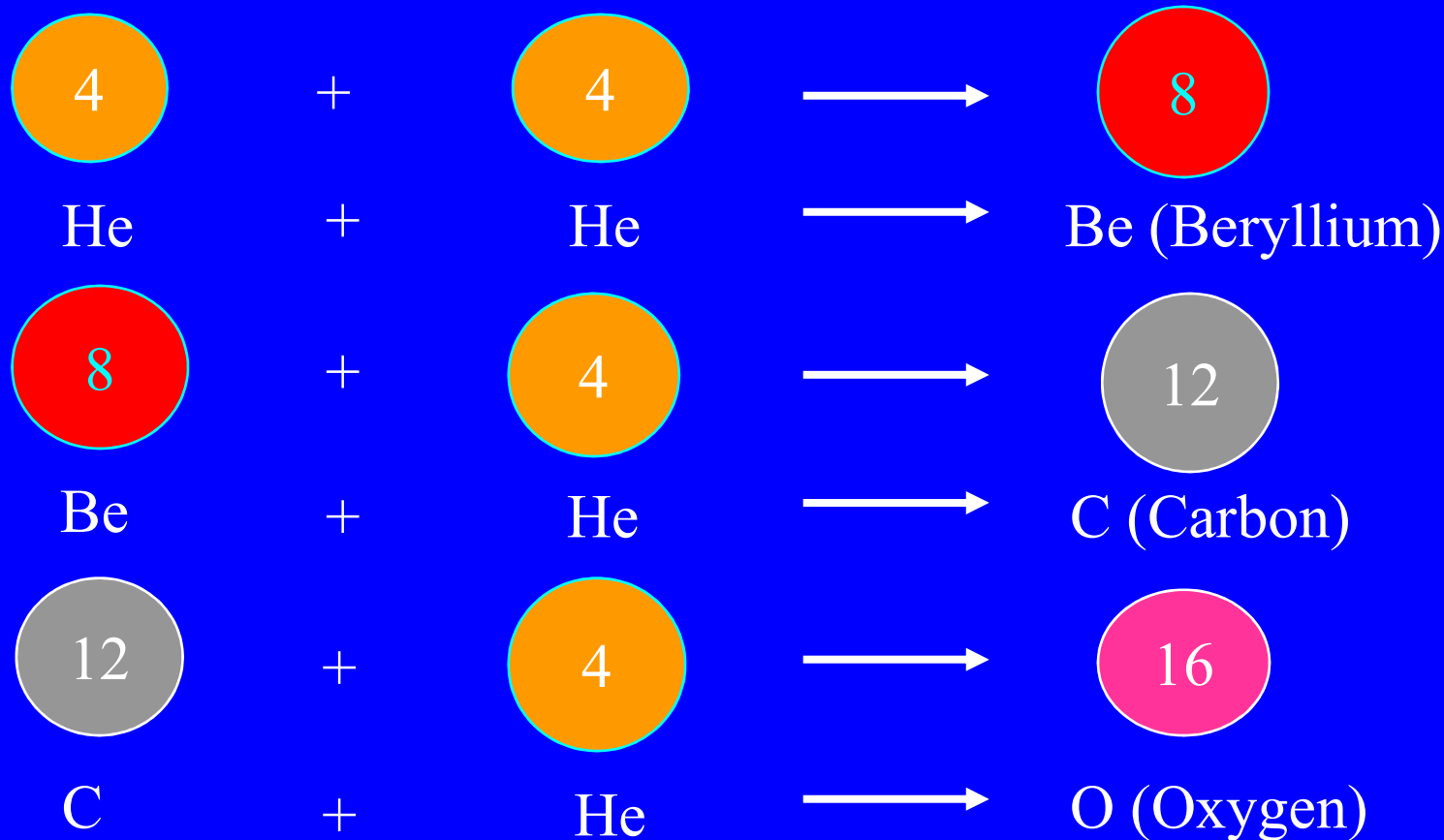


Combine 4 hydrogen atoms to make a helium atom. This is called fusion. It powers the sun.



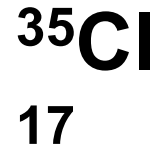
HELIUM ATOM (He)

NUCLEOSYNTHESIS: Making heavier atoms by combining lighter atoms. This occurs in stars and is called fusion. This is where stars get the energy to shine.

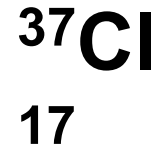


Elements are made in stars.

Isotope Example



chlorine - 35



chlorine - 37

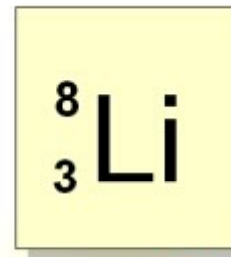
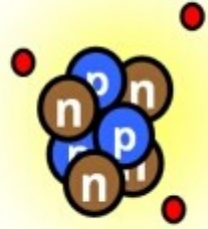
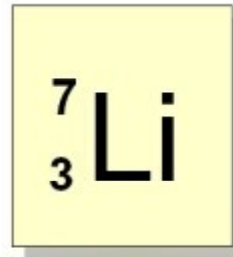
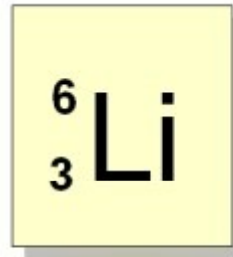
Isotopes of Carbon

Naturally occurring carbon consists of three isotopes, ^{12}C , ^{13}C , and ^{14}C . State the number of protons, neutrons, and electrons in each of these carbon atoms.

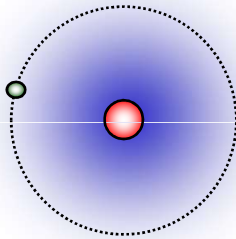
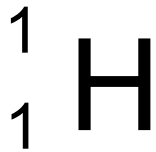
	^{12}C 6	^{13}C 6	^{14}C 6
#P	<u>6</u>	<u>6</u>	<u>6</u>
#N	<u>6</u>	<u>7</u>	<u>8</u>
#E	<u>6</u>	<u>6</u>	<u>6</u>

- Carbon occurs naturally in three isotopes: carbon 12, carbon 13 and carbon 14.
- Every element has its own number of isotopes.
- The addition of even one neutron can dramatically change an isotope's properties.
- Carbon-12 is stable, meaning it never undergoes radioactive decay.
- Carbon-14 is unstable and undergoes radioactive decay with a half-life of about 5,730 years (meaning that half of the material will be gone after 5,730 years).
- This decay means the amount of carbon-14 in an object serves as a clock, showing the object's age in a process called "carbon dating."

Isotopes of Lithium

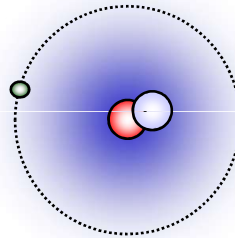
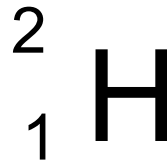


Hydrogen exists as 3 isotopes. Although Hydrogen-1 makes up the vast majority of the naturally occurring element.



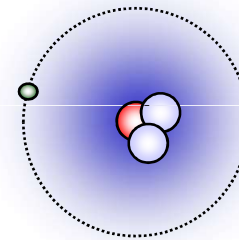
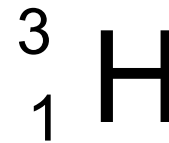
Protons 
Electrons 
Neutrons

Hydrogen



Protons 
Electrons 
Neutrons 

(Deuterium)



Protons 
Electrons 
Neutrons  

(Tritium)

All have a different mass, tritium is radioactive but all react with oxygen to form water.

1. Which of the following is not a sub-atomic particle?

A. Proton.

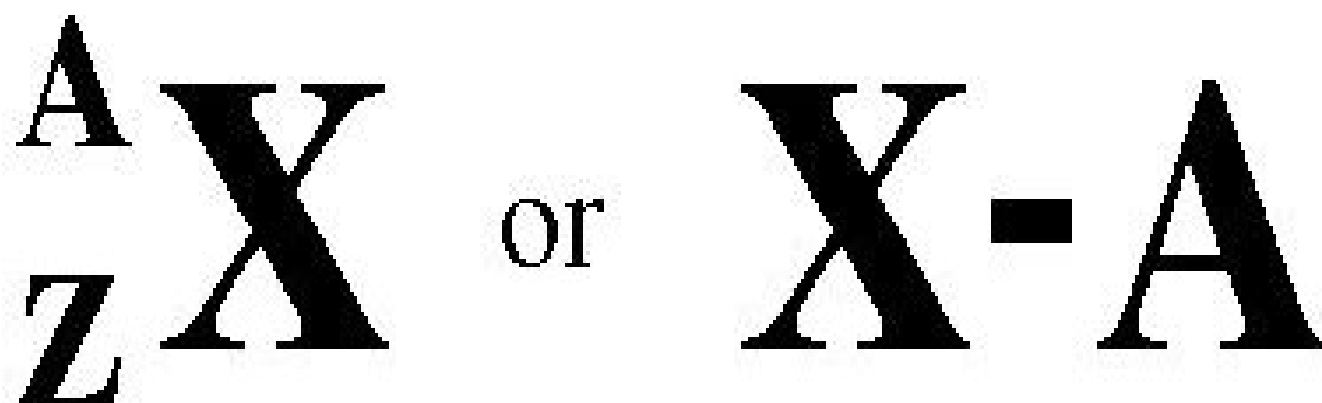
B. Isotope.

C. Neutron.

D. Electron.

Isotope Symbols

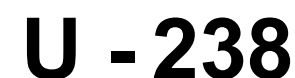
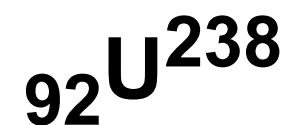
There are two common ways of symbolizing isotopes.



X = element symbol

A = mass number

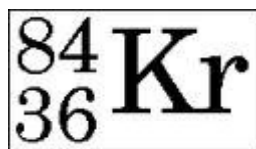
Z = atomic number



Determine the number of neutron in each isotope



1. $238 - 92 = 146$ neutrons



2. $84 - 36 = 48$ neutrons



3. $35 - 17 = 18$ neutrons



4. $14 - 6 = 8$ neutrons

Isotope Facts

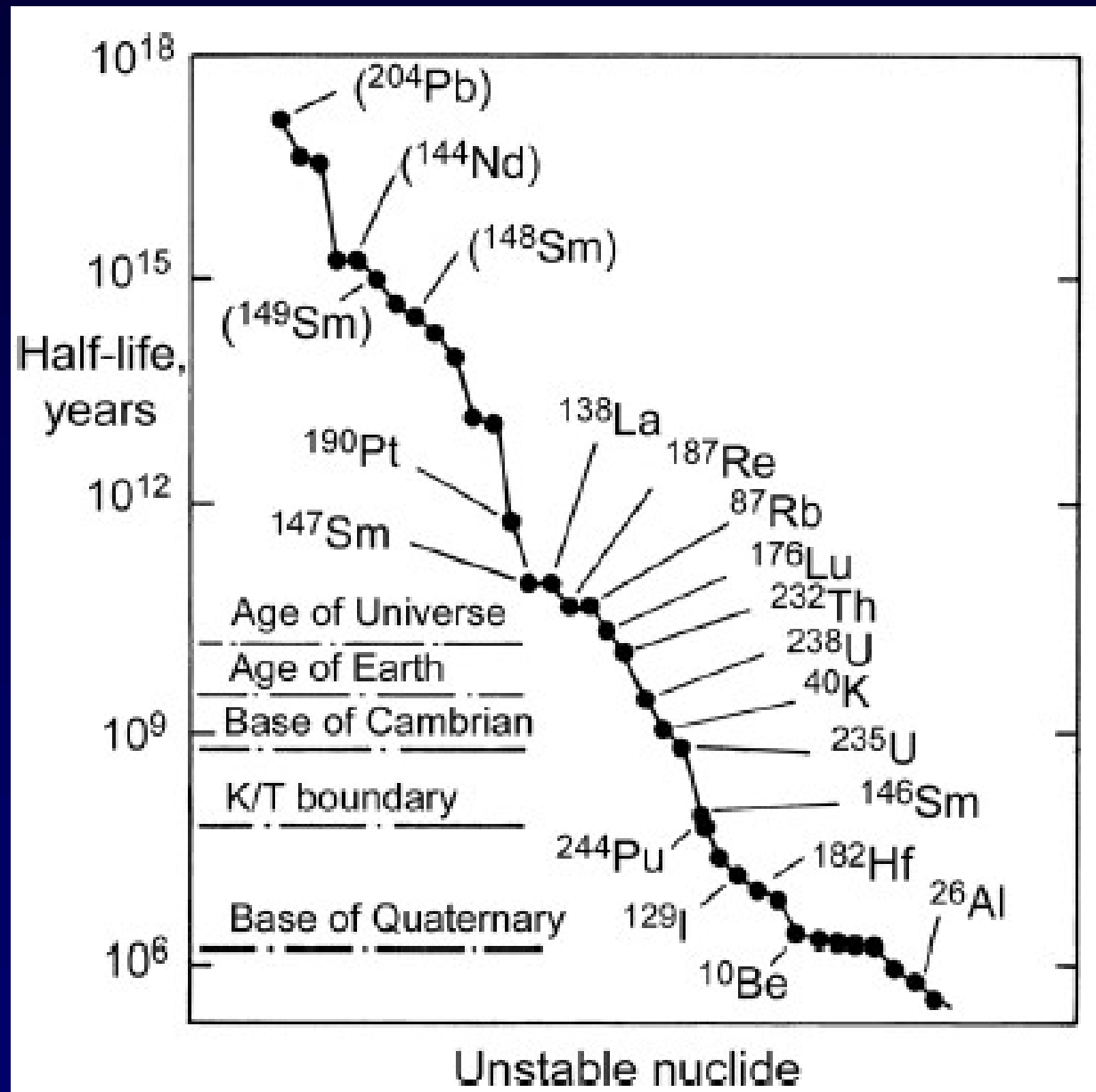
- All elements have isotopes.
- There are two main types of isotopes: stable and unstable (radioactive).
- There are about 254 known stable isotopes, which have not been observed to decay with available detection equipment.
- All artificial (lab-made) isotopes are unstable and therefore radioactive; scientists call them radioisotopes.
- Some elements can only exist in an unstable form (for example, uranium).
- Hydrogen is the only element whose isotopes have unique names: deuterium for hydrogen with one neutron and tritium for hydrogen with two neutrons.

Unstable Nuclides with Half-lives ($t_{1/2}$) over 0.5 Ma

A small number of unstable nuclides have sufficiently long half-lives.

They have not entirely decayed to extinction since the formation of the solar system.

Nuclides with half-lives over 1000 Gy decay too slowly to be geologically useful.



Importance of Isotopes

- Isotopes of an element have the same chemical behavior, but the unstable isotopes undergo spontaneous decay during which they emit radiation and achieve a stable state.
- They are important in nuclear medicine, oil and gas exploration, basic research, and national security.
- Uranium is an example. U-235 and U-238 both contain 92 protons, but U-238 has three extra neutrons and is more stable. Nuclear power reactors work by “enriching” or increasing the percentage of U-235, which is more easily fissioned or split.
- Today, isotopes are a fundamental part of science. Isotopes are used in carbon-14 dating, a method for calculating geological age.
- However, isotopes are not always available in sufficient quantities or at reasonable prices.

Isotones : One or more atoms having an equal number of neutrons are called isotones.

Example: ${}_8\text{O}^{16} \rightarrow P=8, e=8, n=16-8=8$

${}_6\text{C}^{14} \rightarrow P=6, e=6, n=14-6=8$

Example 2

${}_1\text{H}^3 \rightarrow P=1, e=1, n=3-1=2$

${}_2\text{He}^4 \rightarrow P=2, e=2, n=4-2=2$

Isobars: It is derived from the Greek word *isos*, meaning "equal" and *baros*, meaning "weight". Isobars are atoms that have the same mass number.

Example: ${}_{18}\text{Ar}^{40} \rightarrow P=18, e=18, n=40-18=22,$
Atomic mass Number=40

${}_{20}\text{Ca}^{40} \rightarrow P=20, e=20, n=40-20=20$
Atomic mass Number=40

Isoelectronic refers to two atoms, ions, or molecules that have equal number of electrons or "equal charge". Isoelectronic chemical species typically display similar chemical properties.

Example: ${}_{17}\text{Cl}^- \rightarrow P=17, e=18$
 ${}_{18}\text{Ar} \rightarrow P=18, e=18$

Isotonic: In chemistry, we call a solution isotonic when it has the same concentration of the solutes as another solution.

Example: Blood plasma is generally isotonic to the cytoplasmic concentration of blood cells.

The Law of Radioactive Decay

The rate of decay of a radioactive parent nuclide to a stable daughter product is proportional to the number of atoms, n present at any time t (Rutherford and Soddy, 1902):

$$-\frac{dn}{dt} = \lambda n \dots\dots\dots(1)$$

where λ is the constant of proportionality, which is characteristic of the radionuclide in question and is called the decay constant (expressed in units of reciprocal time). The decay constant states the probability that a given atom of the radionuclide will decay within a stated time. The term dn/dt is the rate of change of the number of parent atoms, and is negative because this rate decreases with time. Rearranging equation [1], we obtain:

$$\frac{dn}{dt} = -\lambda n \dots\dots\dots(2)$$

This expression is integrated from $t = 0$ to t , given that the number of atoms present at time $t = 0$ is n_0 .

$$\int_{n_0}^n \frac{dn}{n} = -\lambda \int_{t=0}^t dt \dots\dots\dots(3)$$

Hence $\ln \frac{n}{n_0} = -\lambda t \dots\dots\dots(4)$

Which can be written as $n = n_0 e^{-\lambda t} \dots\dots\dots(5)$

A useful way of referring to the rate of decay of a radionuclide is the 'half-life', $t_{1/2}$, which is the time required for half of the parent atoms to decay. Substituting $n = n_0/2$ and $t = t_{1/2}$ into equation [5], and taking the natural log of both sides, we obtain:

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.69315}{\lambda} \dots\dots\dots(6)$$

The number of radiogenic daughter atoms formed, D^* , is equal to the number of parent atoms consumed:

$$D^* = n_0 - n \dots\dots\dots(7)$$

but $n_0 = n e^{\lambda t}$ (from equation [5]); so substituting for n_0 in equation [7] yields:

$$D^* = n e^{\lambda t} - n \dots\dots\dots(8)$$

$$D^* = n (e^{\lambda t} - 1) \dots\dots\dots(9)$$

If the number of daughter atoms at time $t = 0$ is D_0 , then the total number of daughter atoms after time t is given as:

$$D = D_0 + n(e^{\lambda t} - 1) \dots\dots\dots(10)$$

This equation is the fundamental basis of geochronological dating tools. In the uranium series decay chains, the daughter products of radioactive decay (other than the three Pb isotopes) are themselves radioactive. Hence the rate of decay of such a daughter product is given by the difference between its production rate from the parent and its own decay rate:

$$dn_2/dt = n_1\lambda_1 - n_2\lambda_2 \dots\dots\dots(11)$$

where n_1 and λ_1 are the abundance and decay constant of the parent, and n_2 and λ_2 correspond to the daughter.

But equation [5] can be substituted for n_1 in equation [11] to yield:

$$\frac{dn_2}{dt} = n_{1, \text{initial}} e^{-\lambda_1 t} \lambda_1 - n_2 \lambda_2 \dots\dots\dots(12)$$

this equation is integrated for a chosen set of initial conditions, the simplest of which sets $n_2 = 0$ at $t = 0$. Then:

$$n_2 \lambda_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} n_{1, \text{initial}} (e^{-\lambda_1 t} - e^{-\lambda_2 t}) \dots\dots\dots(13)$$

This type of solution was first demonstrated by Bateman (1910) and is named after him. Recently, Catchen (1984) examined more general initial conditions for these equations, leading to more complex solutions.

Minerals

Rocks containing valuable minerals are called ore. Minerals from ore are used to manufacture products that we use every day. This includes things like houses, stainless steel pots and pans, electronics, batteries, automobiles and fertilizer.

What is Rock ?

A rock is an aggregate of one or more minerals, or a body of undifferentiated mineral matter. Common rocks include igneous rock, sedimentary rock and metamorphic rock.

What is Mineral ?

A mineral is normally an inorganic solid that occurs naturally and has a distinct chemical composition, crystalline form and physical properties. Common minerals include quartz, feldspar, mica, amphibole, olivine, calcite etc.

Minerals are special for five reasons:

1. Minerals are solid,
2. Minerals are mostly inorganic,
3. Minerals are only made by nature,
4. Each mineral is made of only *one* kind of molecule,
5. Minerals have its own orderly internal (crystalline) structure at the atomic level.

Difference between Rocks and Minerals

S. No.	ROCKS	MINERALS
1.	A rock is an inorganic material created spontaneously and has no chemical makeup or atomic structure.	A mineral is similar to a rock, and it is also a solid, inorganic material with a unique crystalline structure and chemical composition.
2.	Minerals are found in rock.	Minerals do not comprise Rocks.
3.	Rocks also exist in a microscopic form.	In nature, they are said to be recognizable.
4.	These are solid minerals found in the Earth's crust.	Mineral deposits occur naturally on the Earth.
5.	Rocks are not edible	Some Minerals are edible, e.g., Salt
6.	Sand, pebbles, shells, and tiny shards of other materials are all examples of rocks.	Minerals include fossil fuels such as coal, petroleum, and others.
7.	Rocks are irregular in form and come in a variety of colors.	Minerals have specific shapes and colors.

Examples of Minerals

Quartz is made up exclusively of oxygen and silicon atoms. More precisely, quartz contains two oxygen (O) atoms and one silicon (Si) atoms. Therefore, the chemical formula for quartz is SiO_2 .

Halite (rock salt) is another mineral whose chemical composition is NaCl . This means that halite is composed of equal numbers of sodium (Na) and chlorine (Cl) atoms. These atoms are arranged in an orderly, three-dimensional lattice that resembles a stack of boxes.

Feldspar a more common mineral whose chemical composition is KAlSi_3O_8 . This reflects not only a more complex composition, but a less symmetric atomic arrangement than that of halite.

Minerals of the Earth's Crust

Name	Chemical Composition	Type of Silicate Structure or Chemical Group
The most common rock-forming minerals (These make up more than 90% of the Earth's crust.)		
<i>Feldspar group</i>		
Plagioclase	Ca and Na Al silicate	Framework silicate
Potassium feldspar (orthoclase, microcline)	K Al silicate	Framework silicate
<i>Pyroxene group</i> (augite most common)	Fe, Mg silicate (some with Al, Na, Ca)	Single-chain silicate
<i>Amphibole group</i> (hornblende most common)	Complex Fe, Mg, Al silicate hydroxide	Double-chain silicate
<i>Quartz</i>	Silica	Framework silicate
<i>Mica group</i>		
Muscovite	K Al silicate hydroxide	Sheet silicate
Biotite	K Fe, Mg Al silicate hydroxide	Sheet silicate

Minerals of the Earth's Crust

Other common rock-forming minerals

Silicates

Olivine	Mg, Fe silicate	Isolated silicate
Garnet group	Complex silicates	Isolated silicate
Clay minerals group	Complex Al silicate hydroxides	Sheet silicate

Nonsilicates

Calcite	CaCO_3	Carbonate
Dolomite	$\text{CaMg}(\text{CO}_3)_2$	Carbonate
Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Sulfate

Much less common minerals of commercial value

Halite	NaCl	Chloride
Diamond	C	Native element
Gold	Au (gold)	Native element
Hematite	Iron oxide (Fe_2O_3)	Oxide
Magnetite	Iron oxide (Fe_3O_4)	Oxide
Chalcopyrite	Cu, Fe sulfide	Sulfide
Sphalerite	Zn sulfide	Sulfide
Galena	Pb sulfide	Sulfide

Physical Properties of Minerals

1. Color: Color is used to identify minerals. Muscovite mica is silvery white or colorless, whereas biotite is black or dark brown. Most of the ferromagnesian minerals (iron / magnesium-bearing), such as augite, hornblende, olivine, and biotite, are either green or black.

Unfortunately, color is also apt to be the most ambiguous of physical properties of minerals. If you look at a number of quartz crystals, for instance, you may find specimens that are white, pink, black, yellow, or purple. Color is extremely variable in quartz and many other minerals because even minute chemical impurities can strongly influence it. Obviously, it is poor procedure to attempt to identify minerals strictly on the basis of color.

2. Streak: A pulverized mineral gives a color, called a **streak**, that usually is more reliable than the color of the specimen itself. Scraping the edge of a mineral sample across an unglazed porcelain plate leaves a streak that may be diagnostic of the mineral. For instance, hematite always leaves a reddish brown streak though the sample may be brown or red or silver.

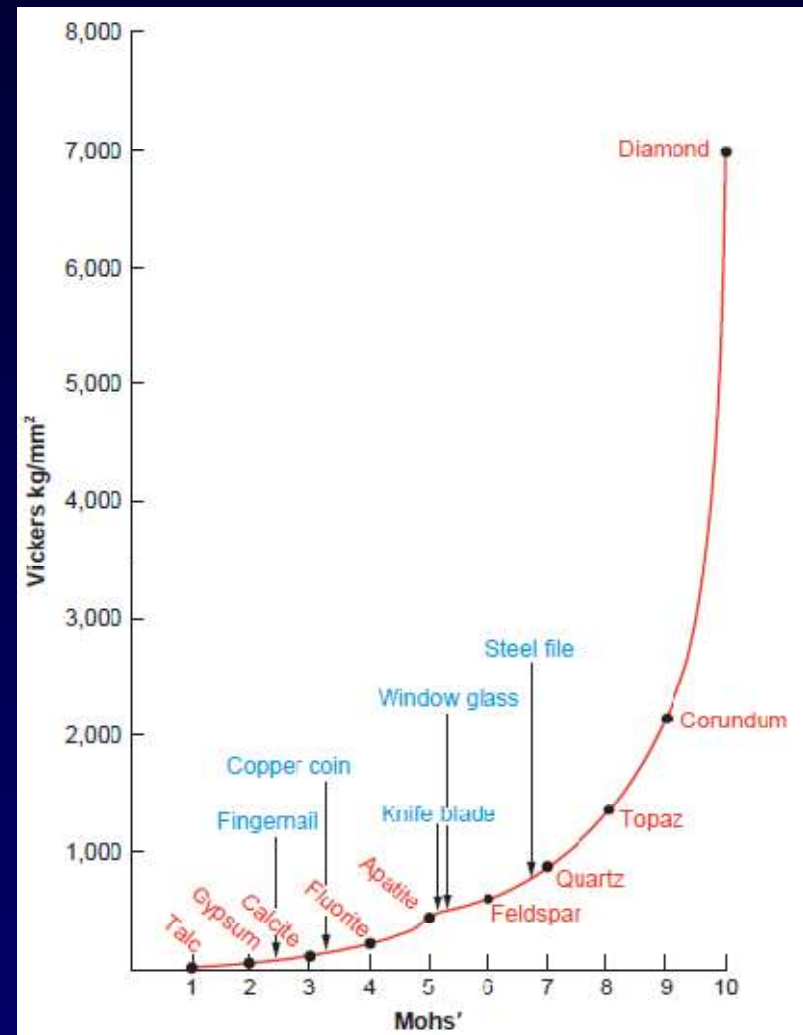
Physical Properties of Minerals *cont....*

3. Luster:

- The quality and intensity of light that is reflected from the surface of a mineral is termed luster. The luster of a mineral is described by comparing it to familiar substances.
- Luster is either metallic or nonmetallic.
- A metallic luster gives a substance the appearance of being made of metal. Metallic luster may be very shiny, like a chrome car part, or less shiny, like the surface of a broken piece of iron.
- Nonmetallic luster is more common. The most important type is glassy (also called vitreous) luster, which gives a substance a glazed appearance, like glass or porcelain. Most silicate minerals have this characteristic. The feldspars, quartz, the micas, and the pyroxenes and amphiboles all have a glassy luster.
- Less common is an earthy luster. This resembles the surface of unglazed pottery and is characteristic of the various clay minerals. Some uncommon lusters include resinous luster (appearance of resin), silky luster, and pearly luster.

Physical Properties of Minerals *cont....*

4. Hardness: The property of “scratchability,” or **hardness**, can be tested fairly reliably. For a true test of hardness, the harder mineral or substance must be able to make a groove or scratch on a smooth, fresh surface of the softer mineral. For example, quartz can always scratch calcite or feldspar and is thus said to be harder than both of these minerals. Substances can be compared to **Mohs’ hardness scale**, in which ten minerals are designated as standards of hardness. The softest mineral, talc, is designated as 1. Diamond, the hardest natural substance on Earth, is 10 on the scale. Mohs’ scale is a relative hardness scale. The absolute hardness is obtained using an instrument that measures how much pressure is required to indent a mineral.



Mohs' hardness scale plotted against Vickers indentation values (kg/mm²). Indentation values are obtained by an instrument that measures the force necessary to make a small indentation into a substance.

Physical Properties of Minerals *cont....*

5. External Crystal Form: The crystal form of a mineral is a set of faces that have a definite geometric relationship to one another. A well-formed crystal of halite, for example, consists of six faces all square and joined at right angles. The crystal form of halite is a cube, in other words.

6. Cleavage: The internal order of a crystal may be expressed externally by crystal faces, or it may be indicated by the mineral's tendency to split apart along certain preferred directions. **Cleavage** is the ability of a mineral to break, when struck or split, along preferred planar directions. A mineral tends to break along certain planes because the bonding between atoms is weaker there. In quartz, the bonds are equally strong in all directions; therefore, quartz has no cleavage. The micas, however, are easily split apart into sheets.

Cleavage is one of the most useful diagnostic tools because it is identical for a given mineral from one sample to another.

Physical Properties of Minerals *cont....*

7. Fracture: Fracture is the way a substance breaks where not controlled by cleavage. Minerals that have no cleavage commonly have an irregular fracture.

Some minerals break along curved fracture surfaces known as conchoidal fractures. Conchoidal fracture is particularly common in glass, including obsidian (volcanic glass).

Minerals that have cleavage can fracture along directions other than that of the cleavage.

8. Specific Gravity: Density is commonly expressed as specific gravity. Specific gravity, also called relative density, is the ratio of the density (mass of a unit volume) of a substance to the density of a given reference material. Most of the common silicate minerals are about two and a half to three times as dense as equal volumes of water. Quartz has a specific gravity of 2.65; the feldspars range from 2.56 to 2.76. Special scales are needed to determine specific gravity precisely. Gold, with a specific gravity of 19.3, is much heavier than galena.

Physical Properties of Minerals *cont....*

9. Transparency: The amount of light able to be passed through a mineral determines its transparency.

(i) **Transparent Minerals:** Light is able to pass through transparent minerals.

(ii) **Translucent Minerals:** partially let the light pass through.

(iii) **Opaque Minerals:** do not let any light pass through.

Flaws, inclusions, and impurities degrade the transparency of a mineral.

10. Tenacity: Tenacity is the resistance of a mineral to breaking, crushing, or bending. Tenacity can be described by the following terms:

- *Brittle* - Breaks or powders easily.
- *Malleable* - can be hammered into thin sheets.
- *Sectile* - can be cut into thin shavings with a knife.
- *Ductile* - bends easily and does not return to its original shape.
- *Flexible* - bends somewhat and does not return to its original shape.
- *Elastic* - bends but does return to its original shape.

Formation of Minerals

Minerals form under an enormously wide variety of conditions - mostly geological; others biological in nature. Some form tens of kilometers beneath the surface; others right at the surface and virtually out of the atmosphere itself. There are three major ways the minerals form.

- Metamorphic minerals form deep underground;
- Igneous minerals form from lava or magma;
- Hydrothermal or sedimentary minerals precipitate from water.
- Minerals can form in many other ways, as well. Some animals manufacture minerals to use as shells. The human body makes its own minerals, too. Bones are made of the mineral apatite.
- The vast majority of minerals are metamorphic, igneous, hydrothermal, or sedimentary.

Formation of Minerals

- The most common minerals are silicates, which are the most abundant elements on Earth. Silicate minerals such as quartz, olivine, and the feldspars (plagioclase and potassium feldspar) crystallize primarily from molten rock (magma).
- Other precipitates include the carbonates calcite and aragonite, which grow in spring and cave waters. Some minerals precipitate due to evaporation (e.g., halite). The very thick salt deposits are found because of the evaporation of seas millions of years ago.
- Some minerals result from biological activity; for example, the building of coral reefs creates huge masses of calcite-rich limestone. Many organisms, including human beings, create magnetite within their skull cases. Bacteria also form huge amounts of sulfur by processing preexisting sulfate minerals.
- Ice may be regarded as a very transient mineral at all but the coldest parts of Earth's surface. (Ice is a major crust-forming mineral on planets of the outer solar system, where it cannot melt).

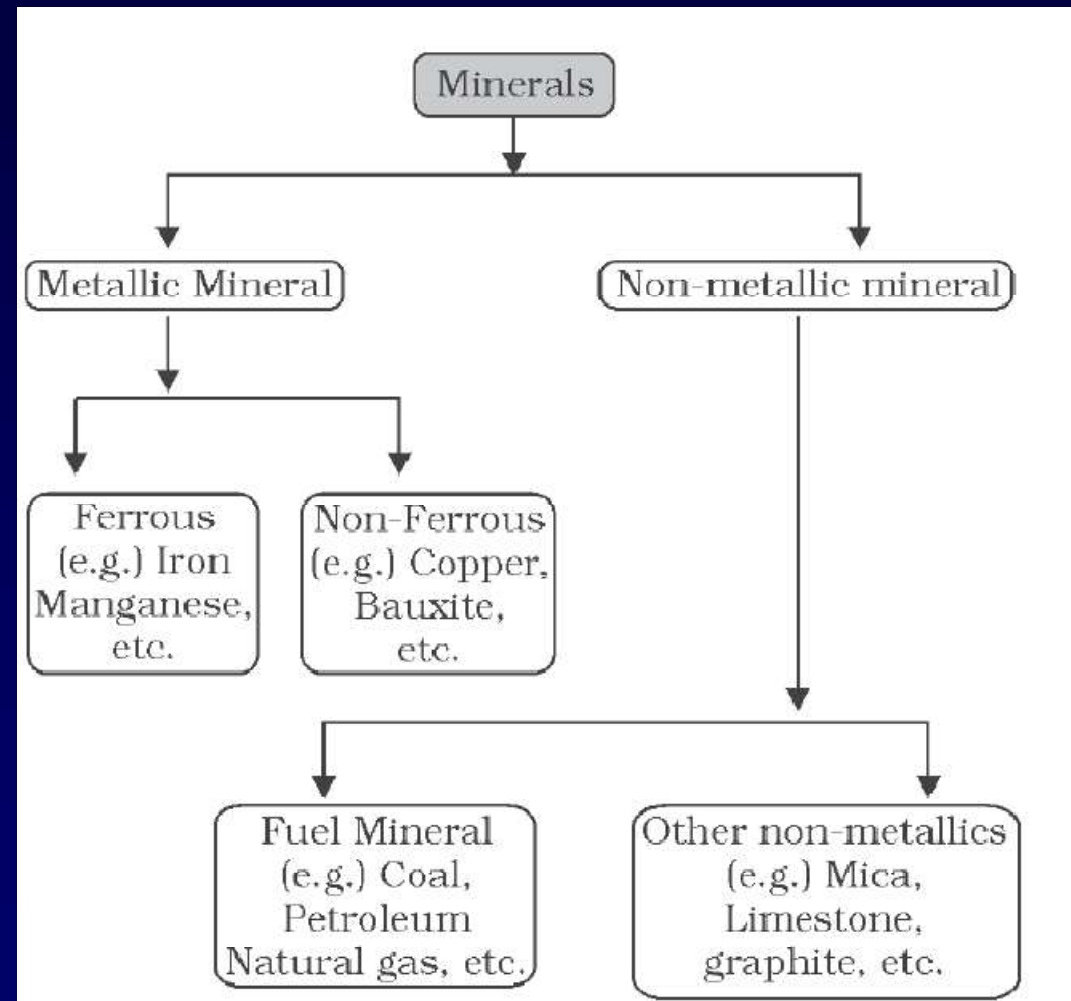
Facts about minerals and its importance

- There are about 4,500 kinds of minerals in the world.
- only a couple hundred are really common.
- only a couple dozen that form the majority of all rocks.
- Each type of mineral is distinguished by a combination of properties, some of which we can see with the unaided eye, others that are discernable only at the microscopic and atomic levels.
- The properties of minerals include color, luster, hardness, chemical composition, and the transmission of light under a microscope etc.

Types of Minerals

On the basis of chemical and physical properties, minerals may be grouped under two main categories of metallic and non-metallic which may further be classified as follows :

- Metallic minerals are the sources of metals.
- Metallic minerals are further divided into ferrous and non-ferrous metallic minerals.
- All those minerals which have iron content are ferrous.
- Those which do not have iron content are non-ferrous.
- Non-metallic minerals are either organic in origin such as fossil fuels also known as mineral fuels which are derived from the buried animal and plant life such as coal and petroleum.
- Other type of non-metallic minerals are inorganic in origin such as mica, limestone and graphite, etc.

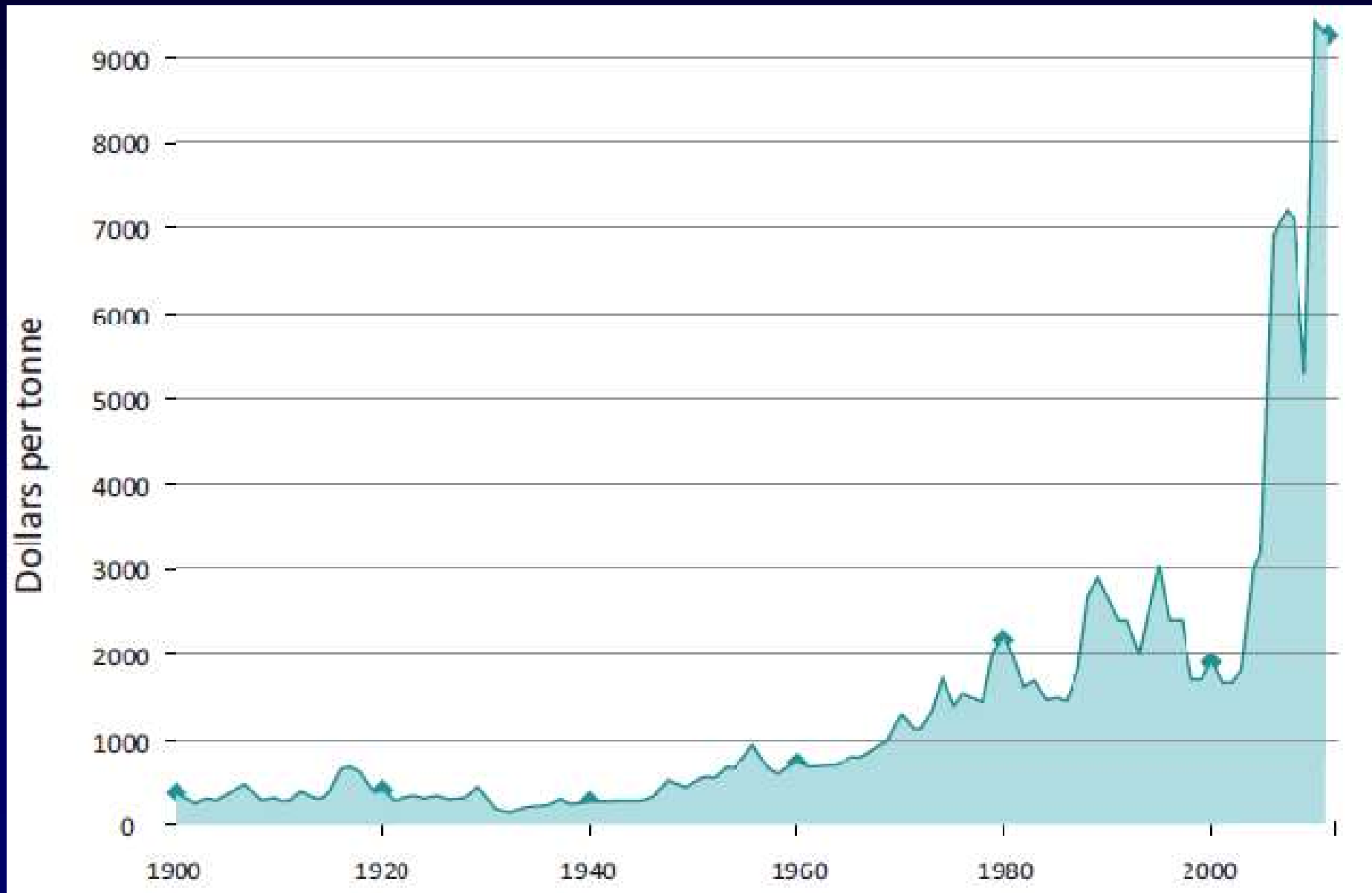


Main types of marine mineral deposits

Type	Location	Exploitable raw material	Depth	Mining status	Economic value
Salt	Coastal	Salt	Inshore	Operational	Moderate
Sand and gravel	Shoreline	Sand and gravel	Shallow	Operational	High
Diamonds	Continental margin	Diamonds	<250 m	Operational	High
Marine placers	Beaches, shallow waters	Tin, gold, chromium, zirconium, rare earths, titanium	Shallow	Operational	Moderate
Phosphates	Shallow waters and former volcanoes	Phosphate	Shallow to moderate depth	Non-operational	Low
Nodules	Large abyssal plains	Copper, cobalt, nickel	4,500 to 5,500 m	Potential resources	Moderate
Manganese crust	Former intraplate volcanoes	Cobalt, copper, platinum	1,000 to 2,500 m	Potential resources	Moderate to high
Hydrothermal sulphides	Volcanic ridges	Copper, zinc, silver, gold, cobalt, lead	1,000 to 4,000 m	Potential resources	High

Marine mineral can be divided into (i) shallow-water mineral deposits (< 200 m), which are already being mined, and (ii) deep-sea mineral deposits (< 1000 m), whose potential is attracting increasing interest from mining companies.

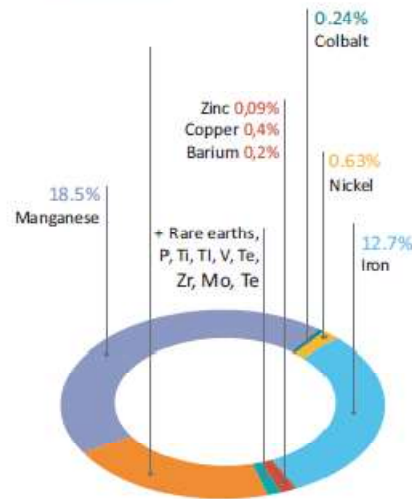
Copper prices during the twentieth century



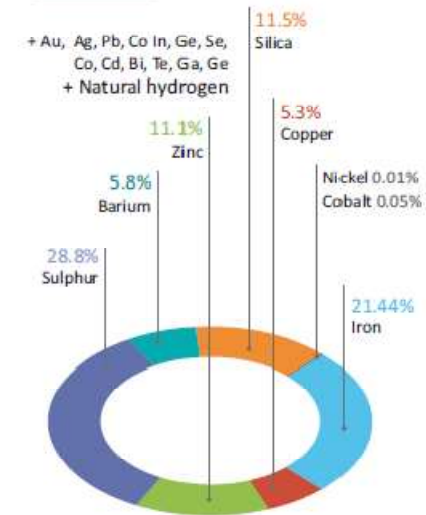
Concentrations of major elements in deep water mineral deposits



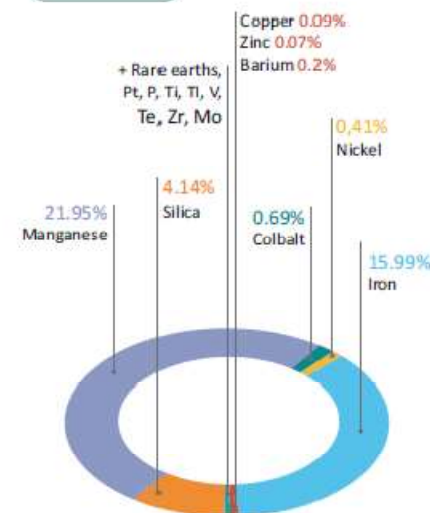
Nodules



+ Au, Ag, Pb, Co, In, Ge, Se, Co, Cd, Bi, Te, Ga, Ge + Natural hydrogen



Crusts



Placer Deposits

- Minerals that form placer deposits have high specific gravity, are chemically resistant to weathering, and are durable. Such minerals include gold, platinum, cassiterite, magnetite, chromite, ilmenite, rutile, native copper, zircon, monazite, and various gemstones.
- There are several varieties of placer deposits:
 - (i) stream, or alluvial placers; (ii) eluvial placers; (iii) beach placers; and (iv) eolian placers.

Stream placers or alluvial placers depend on swiftly flowing water for their concentration.

Eolian placers may form in arid areas where wind, not water, acts as the concentrating agent, removing fine particles of the lighter dross. The gold deposits of some parts of the Australian desert are examples.

Eluvial placers form on hill slopes from weathered deposits. They are not acted on by streams but by rainfall and wind, which carry away the light materials. Thus they may be considered intermediate in the formation of stream placers. Examples include the earlier worked gold deposits of Australia and the cassiterite placers of Malaysia.

Beach placers form on seashores where wave action and shore currents shift materials.

The lighter materials are shifted more rapidly than the heavier, thus concentrating them.

The placer mineral deposits along the east coast of India are known for the economic concentration of ilmenite, sillimanite, zircon, monazite, etc.

Indian resources constitute about 35% of world resources of ilmenite, 10% of rutile, 14% of zircon and 71.4% of monazite.

Among the examples of beach placers are the gold deposits of Nome, Alaska; the zircon sands of Brazil and Australia; the black sands (magnetite) of Oregon and California; and the diamond-bearing marine gravels of Namaqualand, South Africa.

Heavy mineral is a mineral with a density that is greater than 2.9 g/cm^3 , most commonly referring to dense components of siliciclastic sediment.

Heavy mineral sands are a class of ore deposit which is an important source of zirconium, titanium, thorium, tungsten, rare-earth elements, the industrial minerals diamond, sapphire, garnet, and occasionally precious metals or gemstones.

Heavy mineral sands are placer deposits formed most usually in beach environments by concentration due to the specific gravity of the mineral grains.

Factors Affecting Mineral Availability

In the medium and long term (over ten years), availability is related to five factors:

- Geological (Does the resource exist?)
- Technical (Can the ore be extracted and processed?)
- Environmental and social (Can metals be produced in an ecologically and socially acceptable manner?)
- Political (How can governments influence availability through their actions?)
- Economic (Can metals be produced at a cost that users are prepared to pay?)



Thank You

Pillow lava at CIR observed during manned submersible dive